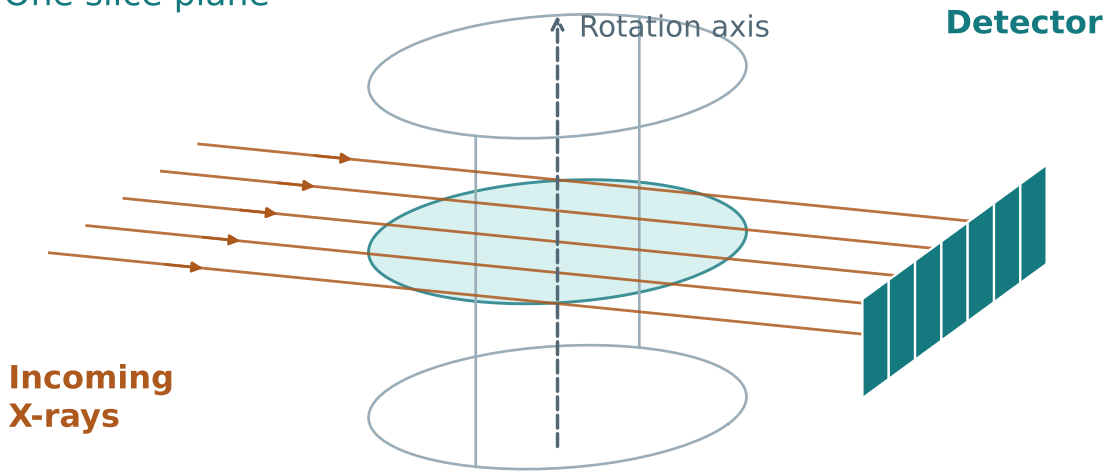
How detector rows sample a CT slice

One view of an ideal parallel-beam scan

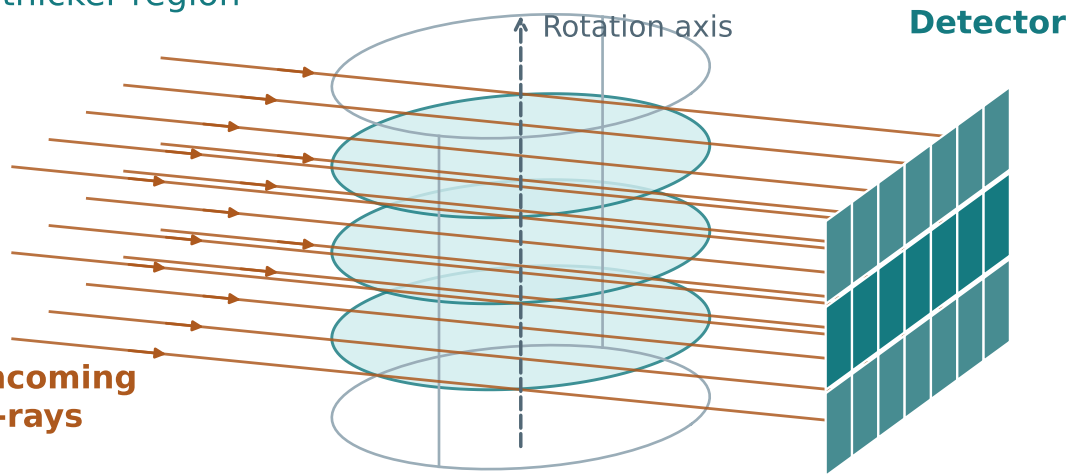
One detector row

One slice plane



Multiple detector rows

A thicker region



The rays travel within the slice plane. As the view rotates, the rays change direction within that same plane.

The right panel extends the parallel-beam model. Clinical multirow CT uses a beam that diverges across detector rows.